

Self-Balancing Robot: Component-Wise Inertia Analysis

Dynamics Calculation

January 18, 2026

1 Physical Parameters

Global Parameter	Symbol	Value
Gravity	g	9.81 m/s ²
Total Body Mass	m_{total}	0.89 kg
Global Center of Mass (from axle)	L	0.086 m
Wheel Radius	R	0.0325 m
Wheel Mass	M	0.05 kg

2 Detailed Inertia Calculation (J_c)

To obtain the precise Pitch Inertia about the Center of Mass (J_c), we sum the contributions of every part using the **Parallel Axis Theorem**:

$$J_c = \sum_{i=1}^n (I_{local,i} + m_i(h_i - L)^2)$$

Where h_i is the height of that component from the axle, and L is the global COM (0.086m).

2.1 Component Definition

We decompose the 890g (0.89kg) body into 4 parts:

1. **Base Assembly (Motors + 5cm Extrusion):** Located at the bottom.
2. **Vertical Mast (25cm Extrusion):** The main spine of the robot.
3. **Crossbar (10cm Extrusion):** Structural support.
4. **Battery Holder (4S LiPo):** The heaviest dense mass.

2.2 Summation Table

Table 1: Inertia Summation

Component	Mass (m_i) (kg)	Height (h_i) (m)	Dist to COM (d) $ h_i - 0.086 $	Local I_{loc} (10 ⁻⁴ kg·m ²)	Shift $m_i d^2$ (10 ⁻⁴ kg·m ²)
1. Base (Motors + 5cm)	0.300	0.030	0.056	0.60	9.41
2. Mast (25cm Extrude)	0.140	0.125	0.039	7.29	2.13
3. Crossbar (10cm Ext)	0.050	0.180	0.094	0.04	4.42
4. Battery (4S)	0.400	0.090	0.004	3.50	0.06
Sum	0.890			11.43	16.02

Calculation of J_c :

$$J_c = \sum I_{loc} + \sum m_i d^2$$

$$J_c = (11.43 \times 10^{-4}) + (16.02 \times 10^{-4})$$

$$J_c = 27.45 \times 10^{-4} \text{ kg} \cdot \text{m}^2$$

$$J_c = 0.00275 \text{ kg} \cdot \text{m}^2$$

3 System Constants

3.1 Numerator (Num)

$$\text{Num} = 2M + m + \frac{2J_w}{R^2}$$

$$\text{Wheel Inertia } J_w = 0.5 \times 0.05 \times 0.0325^2 = 0.000026.$$

$$\text{Num} = (2 \times 0.05) + 0.89 + \frac{2(0.000026)}{0.0325^2}$$

$$\text{Num} = 0.1 + 0.89 + 0.05 = \mathbf{1.04}$$

3.2 Denominator (den)

Using the calculated $J_c = 0.00275$:

$$\text{den} = -m^2 L^2 + (mL^2 + J_c)(\text{Num})$$

$$m^2 L^2 = (0.89 \times 0.086)^2 = 0.00586$$

$$mL^2 + J_c = (0.89 \times 0.086^2) + 0.00275 = 0.00658 + 0.00275 = 0.00933$$

$$\text{den} = -0.00586 + (0.00933 \times 1.04)$$

$$\text{den} = -0.00586 + 0.00970$$

$$\text{den} = \mathbf{0.00384}$$

4 Final State-Space Matrices

With $\text{den} = 0.00384$:

4.1 Matrix A

$$A_{3,2} = \frac{-m^2 L^2 g}{\text{den}} = \frac{-0.00586 \times 9.81}{0.00384} = \mathbf{-14.96}$$

$$A_{4,2} = \frac{mgL \cdot \text{Num}}{\text{den}} = \frac{0.89 \times 9.81 \times 0.086 \times 1.04}{0.00384} = \mathbf{203.35}$$

$$\mathbf{A} = \begin{bmatrix} 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \\ 0 & -14.96 & 0 & 0 \\ 0 & 203.35 & 0 & 0 \end{bmatrix}$$

4.2 Matrix B

$$B_3 = \frac{-mL}{\text{den}} = \frac{-0.89 \times 0.086}{0.00384} = \mathbf{-19.93}$$

$$B_4 = \frac{\text{Num}}{\text{den}} = \frac{1.04}{0.00384} = \mathbf{270.83}$$

$$\mathbf{B} = \begin{bmatrix} 0 \\ 0 \\ -19.93 \\ 270.83 \end{bmatrix}$$